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Imaging of electrical response of NiO_x under controlled environment with sub-25-nm resolution

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Abstract. The spatially resolved electrical response of polycrystalline NiO_x films, composed of 40 nm crystallites, was investigated under different relative humidity (RH) levels. The topological and electrical properties (surface potential and resistance) were characterized with sub-25-nm resolution using Kelvin probe force microscopy and conductive scanning probe microscopy under argon atmosphere with 0%, 50%, and 80% RH. The dimensionality of surface features obtained through autocorrelation analysis of topological maps increased linearly with increased RH, as water was adsorbed onto the film surface. Surface potential decreased from 280 to 100 mV and resistance decreased from 5 GΩ to 3 GΩ, in a nonlinear fashion when RH was increased from 0% to 80%. Spatially resolved surface potential and resistance of the NiO_x films was found to be heterogeneous throughout the film, with distinct surface features that grew in size from 60 to 175 nm at 0% and 80% RH levels, respectively. The heterogeneous character of the topological, surface potential, and resistance properties of the polycrystalline NiO_x film observed under dry conditions decreased with increased RH, yielding nearly homogeneous surface properties at 80% RH, suggesting that the nanoscale potential and resistance properties converge with the mesoscale properties as water is adsorbed onto the NiO_x film. © *2016 Society of Photo-Optical Instrumentation Engineers (SPIE)* [DOI: 10.1117/1.JPE.6.038001]

Keywords: nickel oxide; humidity; local conductivity; Kelvin probe force microscopy; conductive atomic force microscopy.

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1 Introduction

N-type metal oxides, (SnO_2, ZnO, and TiO_2) and p-type metal oxides (NiO_x, Cr_2O_3) are promising selective electron and hole transport electrode materials for photovoltaic devices. Such metal oxides demonstrate surface- and environment-dependent electrical behavior, which is attributed to changes to the conduction band energy, related to electron or hole injection from surface adsorbed molecules; however, the majority of research elucidating the underlying mechanisms has primarily centered on n-type metal oxides single crystals.[1,2] Consequently, spatially resolved chemiresistive response of p-type metal oxide semiconductors at the mesoscale is still in the early stages of development, and the spatially resolved resistive response of p-type metal oxides is not yet well understood.[3,4]

The surface of polycrystalline NiO_x films typically consists of either stable NiO_x (1 0 0) planes, or polar NiO_x (1 1 1) planes, which differ markedly in reactivity to oxygen and water adsorption.[5,6] Oxide surfaces are generally hydroxylated to varying extents in ambient conditions, and the presence of hydroxyls at the NiO_x surface stabilizes the thermodynamically unfavored (1 1 1) surface.[7–9] Using spectroscopy techniques, it has been shown that the polar

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characteristics, and the easy accessibility of Ni^{2+} surface sites readily allow gas adsorption at the NiO_x (1 1 1) surface, yielding the formation of hydroxyl surface groups, whereas nonpolar NiO_x (1 0 0) planes resist oxygen adsorbtion, inhibiting the formation of hydroxyl surface groups. Water vapor at the NiO_x (1 1 1) surface undergoes dissociative adsorption leading to the formation of hydroxyl groups on the NiO_x surface. At higher humidity levels, the NiO_x surface can be saturated with hydroxyl groups and subsequent molecules of water are adsorbed to the hydroxyl groups via hydrogen bonding.[10,11] In films with mixed grain surfaces, exposure of the film to water may lead to reorientation of NiO_x (1 0 0) surface, resulting in growth of NiO_x (1 1 1) grain size.[12–14]

The electrically controlled redox state of Ni in NiO_x films has found utility in electrochromic and data storage devices. P-type NiO_x has been shown to be an effective hole extraction layer in bulk heterojunction solar cells, and an electron blocking/hole transport layer in organic and hybrid photovoltaic cells (OPVs).[15–18] The incorporation of NiO_x in OPVs, instead of conventional PeDOT:PSS, improved the device stability in humid environments.

While the physical mechanism linking the electrical response to water adsorption on NiO_x surfaces is known from studies on single crystal and epitaxial NiO_x thin-films, the majority of real-world applications would utilize polycrystalline thin-films or nanoparticle assemblies; therefore, spatially resolved study of adsorption on the scale of single nanoparticles and nanoparticle assemblies promises to advance a broad range of applications including photovoltaic devices, catalysis, and sensors. Here, we correlate topological changes in the polycrystalline NiO_x surface with spatially resolved surface potential and resistive responses to increase RH levels, with a spatial resolution of less than 25 nm. Correlation of the spatially resolved surface potential and resistance response values to area averaged values links the nanoscale properties to the mesoscale properties of the polycrystalline NiO_x film.

2 Methods

2.1 *NiO_x Film Synthesis*

Stoichiometric NiO_x targets were used to deposit NiO_x films on room temperature Si substrates by radio frequency-magnetron sputtering using an AJA International, Inc. sputtering system. NiO_x films were deposited on precleaned boron doped (1 0 0) Si substrates. The chamber was evacuated, and then refilled to 9 mTorr with an argon/oxygen mixture (80% Ar and 20% O_2) and sputtering was performed at 350 W, resulting in a deposition rate of 12.7 Å/min. After deposition, the NiO_x film was annealed in air by 25 pulses (500 μs/pulse) from a Xe lamp [PulseForge 3300 pulsed thermal processing (PTP) system] operating at 300 V with a peak power of 4.3 (kW/cm^2). Prior to characterization, excess surface water was removed and the surfaces were cleaned and activated in vacuum by UV irradiation from a Hg lamp for 12 h, and were allowed to equilibrate under vacuum for another 12 h. The synthesis is consistent with previously used methods that result in nonstoichiometric NiO_x films.[5,19,20] Synthesis of NiO_x in excess oxygen results in interstitial oxygen, and pulsed thermal annealing and UV irradiation results in diffusion and partial removal of interstitial oxygen, leading to partial recovery of NiO stoichiometry.[20] However, pulsed annealing does not completely recover the stoichiometry, and thus the resulting NiO_x film composition is considered to be nonstoichiometric.

2.2 *Characterization*

X-ray diffraction (XRD) was measured on a PANalytical X'Pert Pro MPD diffractometer with a CuKa source, Ni filter, and X'Celerator detector. Scanning electron microscope (SEM) images were taken with a ZEISS Merlin SEM with a secondary electron detector using an accelerating voltage of 5 kV. Topological and surface potential measurements were recorded using a scanning probe microscope (SPM), Asylum Research Cypher S, with a conductive Pt/Ir coated cantilever operated at 75 MHz resonance frequency. During Kelvin probe force microscopy (KPFM) scans, the tip was first piezoelectrically scanned over the sample at the resonance frequency in noncontact mode to measure the topology, and then scanned over the sample again but was driven with an alternating current, forming a capacitor with the sample surface. During the second scan

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(the KPFM scan) a potential difference between the sample and the tip causes the cantilever to vibrate, and a null circuit applies a DC voltage to the tip to minimize the vibration. The magnitude of the applied DC voltage is equal to the contact potential difference between the cantilever and the sample (V_{cpd}). Presuming the surface potential of the tip remains constant, the surface potential of the sample can be determined by $V_{cpd} = V_{tip} - V_{sample}$.[21–23] Resistance maps were recorded by placing the tip in contact with the sample surface and applying a constant voltage bias (+4.2 and −4.2 V) to the tip. The tip was then retracted from the surface, moved to the next measurement location to diminish any electric field built up between measurements, and then driven back into contact with the surface for measurement. Spatially resolved resistance was determined from the resultant current measured at each point. The spatially resolved electrical resistance of the NiO_x film was mapped at 4096 individual points over a 1 μm^2 area (64×64 points map), and multiple areas were scanned to ensure reproducibility across the sample. Feature size was determined from the SPM maps using radial autocorrelation analysis, which gives a point-to-point signal correlation value and compares the correlation value to the distance between points. The correlation value minimum is indicative of the feature size.

3 Results and Discussion

3.1 *Film Characterization*

SEM images of the film surface show that the NiO_x film is composed of interconnected polycrystalline nanoparticle domains of relatively uniform size [Fig. 1(a)]. The SEM image of the film cross section [Fig. 1(b)] shows the NiO_x layer is 170-nm thick and consists of dense 20 to 50 nm columnar polycrystalline structures. The crystalline structure of the sputtered NiO_x film [Fig. 1(a) (inset)] is polycrystalline cubic phase NiO_x with a preferred (1 1 1) orientation, and an average crystallite size of 43 nm. The NiO_x (1 1 1) structure denotes a film primarily composed of alternating layers of oxygen ions and nickel ions, resulting in a polar surface terminated by dangling bonds. These dangling bonds are readily stabilized by adsorption of oxygen or formation of hydroxyl groups when exposed to air or moisture.[11,13]

The topology of the NiO_x film surface was measured using atomic force microscopy (AFM) in tapping mode, illustrated by the schematic in Fig. 1(c). The scans were performed on representative 1 μm^2 areas of the NiO_x film under argon atmosphere with 0%, 50%, and 80% relative humidity (RH). RH was determined by the following equation:

$$RH = \frac{\rho_{H_2O}}{\rho^*_{H_2O}} \times 100\%,$$

where ρ_{H_2O} represents the partial pressure of water vapor in the environment, and $\rho^*_{H_2O}$ is the equilibrium water vapor pressure at a given temperature of 23°C.

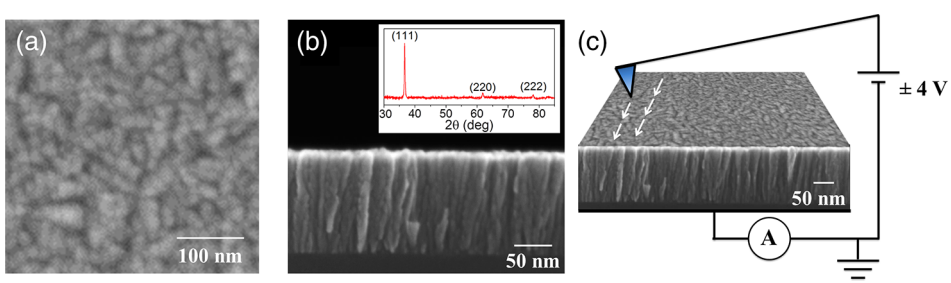

Fig. 1 (a) A SEM micrograph of the NiO_x film shows a surface comprised of distinct crystallite domains and (b) cross-section SEM shows the 170-nm thick film is composed tall columnar structures of 20–50 nm in diameter. XRD of the NiO_x films describes a polycrystalline cubic structure with preferred (1 1 1) orientation (inset). Schematic of the experimental setup (c) used to measure topology and surface potential, under KPFM mode in noncontact mode, and to determine localized resistance through the NiO_x film. Resistance was measured by applying either a positive or negative bias while in contact mode, where the potential was applied to the SPM tip and current through the film was measured. The localized current was measured each of the 4096 points by applying a DC voltage bias to the tip with the current amplifier grounded.

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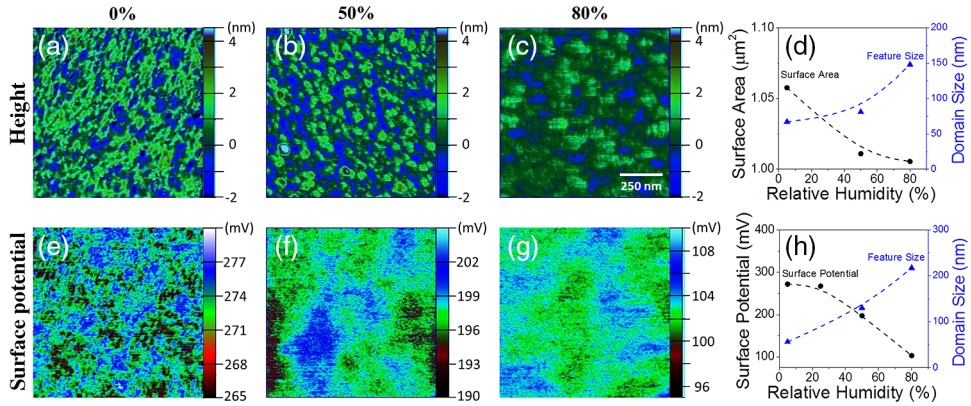

Fig. 2 AFM topology images of a 1 μm^2 area of a NiO_x film at (a) 0%, (b) 50%, and (c) 80% RH show a decrease in surface roughness, and (d) an increase in topological feature size with increased RH. KPFM maps of surface potential at (e) 0%, (f) 50%, and (g) 80% RH show an overall decrease in the average surface potential, and (h) an increase in surface potential features sizes with increasing RH.

The topological maps [Figs. 2(a)–2(c)] show that under dry conditions the surface feature height is 6 nm (roughness due to NiO_x nanocrystals) with a lateral feature size of 67 nm and an estimated surface area of 1.06 μm^2. At 50% RH, the surface roughness decreased to 5.5 nm, while the feature size increased to 84 nm, with an estimated surface area of 1.01 μm^2. The surface roughness further decreased to 2 nm at 80% RH, as the feature size increased to 140 nm with an estimated surface area of 1.00 μm^2. Figure 2(d) (black circles) displays the surface area and feature size as a function of increasing humidity, and the connecting splines reveal a general decrease in surface area and increase in feature size.

Uniform adsorption of water is indicative of the polar surface character similar to that of the NiO_x (1 1 1) surface.[11,24,25] Kitakatsu et al.[12] reported the possibility of structural reorientation of NiO_x (1 0 0) to hydroxylated NiO_x (1 1 1) surfaces resulting from water adsorption, leading to the growth in diameter of NiO_x (1 1 1) grains. However, such a structural reorientation required thermal annealing to remove surface hydroxyls and subsequent water exposure to induce the transformation. An alternative mechanism may involve adsorption of multilayers of water on the rough NiO_x surface, which would lead to the observed increase in surface feature size and reduction of surface roughness. Figure 3 illustrates how this hypothesis could explain the changes in topography of the NiO_x surface observed as RH increases. As RH increases,

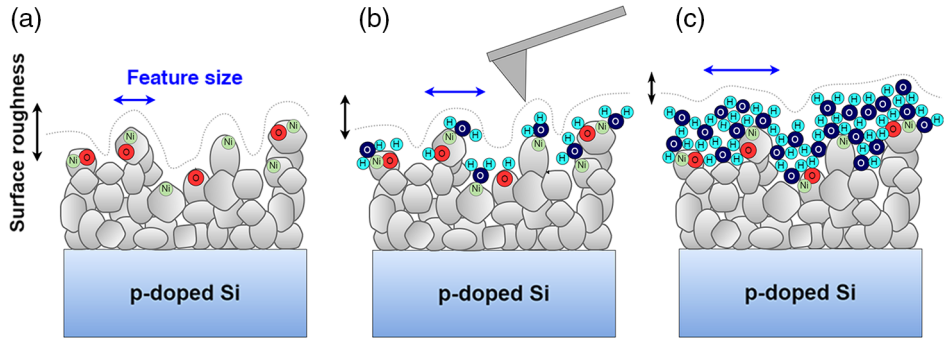

Fig. 3 Illustration depicting the relationship between surface roughness and water adsorption in a polycrystalline NiO_x film with RH increasing from left to right. At low RH values (a) few hydroxyl groups or water molecules inhabit the NiO_x surface resulting in a varied surface profile with relatively high roughness. Under increased RH (b) water and oxygen adsorb onto the surface, decreasing the roughness and increasing the size of the dimensionality of surface features, and at high RH (c) a monolayer of water coats the NiO_x surface, and hydrogen bonding extends the thickness of the water layer, further decreasing the surface roughness, and surface feature size.

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and water is adsorbed onto the NiO_x surface, crevices between the nanocrystals become filled with water molecules, leading to decreased surface roughness. Concurrently, water adsorbed on the polycrystalline NiO_x surfaces increases the size of the topological features. Given that hydroxylation of the NiO_x surface can lead to ordered hydrogen bonding of water, multiple layers of water are likely adsorbed to the crystalline surface at high RH values, resulting in the decreased roughness and increased feature size observed.[12]

The change in surface potential of NiO_x under increasing RH level was characterized using KPFM, and each sample area was scanned multiple times to ensure that the measured results were reproducible [Figs. 2(e)–2(g)]. At 0% RH, the size of surface potential features is 56 nm, similar to the feature size obtained from topologic maps. In dry conditions, the surface potential maps show that the surface potential varies across the film, and is segregated into 50 nm regions of similar surface potential. In increased RH, the surface potential feature sizes increased to 140 nm at 50% and 210 nm at 80% RH, respectively. Figure 2(h) shows that the average surface potential value decreases with increasing RH, from 280 mV at 0% to 100 mV at 80%. Adsorbed water smooths the heterogeneous character of the NiO_x surface potential, yielding a decrease in an average value of surface potential and larger size of surface potential features, suggesting a screening effect NiO_x by hydroxyl groups and subsequent molecular water adsorption on the NiO_x surface. Water adsorption has been shown to linearly increase the work function of NiO_x, which correlates well with our observation of decreased surface potential in the presence of water, since an increase in the value of surface potential is proportional to a decrease in work function.[24,25] The nonuniformity of surface potential observed at high humidity suggests that chemisorption may not be even across the polycrystalline film surface, with some adsorption sites being more active than others. This is likely caused by the heterogeneous nature of the polycrystalline NiO_x film where both (1 0 0) and (1 1 1) terminated crystallites are exposed to environment. The heterogeneous character of the spatially resolved surface potential at low humidity levels contrasts with the almost homogeneous character of surface potential at high RH levels, suggesting that at high RH levels the NiO_x properties at the nanoscale and at the mesoscale converge.

Figures 4(a)–4(c) depict the spatially resolved resistance was measured under 0%, 50%, and 80% RH at +4.2 V, and Figs. 4(e)–4(g) shows the localized resistance at −4.2 V. At 0% RH, the resistance maps at positive and negative biases are similar with average resistance values of 4.5 and 4.9 GΩ, respectively, and similar nanosized conducting features of 100 nm. The size of

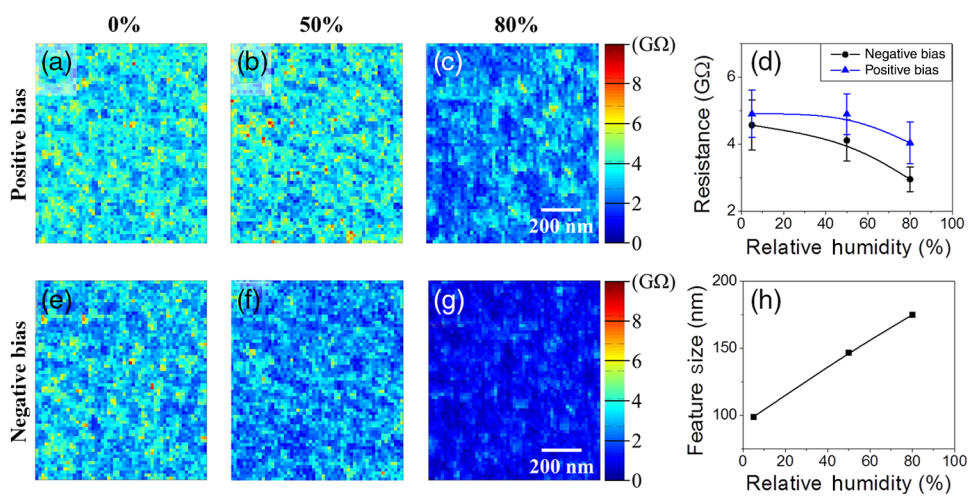

Fig. 4 False color resistance maps of a NiO_x film at (a, b, c, respectively) 0%, 50%, and 80% RH under positive bias (d, e, f, respectively) and under negative bias. Resistance maps are composed of 4096 individual points across the NiO_x film, calculated from the current measured while at +4.2 or −4.2 V bias. The maps reveal an overall decrease in average resistance with increasing humidity, but show consistently lower resistance values at negative potentials. Averaged resistance values reflect this decrease (g), and suggest that high humidity levels have a greater impact at negative potentials. (h) Conducting feature size increases with increasing RH levels.

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the conducting features estimated by a radial autocorrelation function was found to be similar at either positive or negative bias, increasing in size from 100 nm in 0% RH to 175 nm at 80% RH [Fig. 3(h)]. The average resistance decreased with increasing RH levels at both positive and negative biases [Fig. 3(d)]. Heterogeneity of spatially resolved resistance at −4.2 V is lower than that at +4.2 V. At positive bias, the nanoscale and mesoscale properties of NiO_x have not converged at 80% RH, whereas at negative bias, a more uniform distribution of spatially resolved resistances was observed. The conducting feature size determined from the resistance maps correlate well with the feature size obtained from surface potential and topological maps, suggesting that all the measurements were done on the same characteristic features of polycrystalline NiO_x.

To validate the locally measured resistance of the NiO_x film, the through-film macroscale resistivity was determined from Hall effect measurements and found to be 17.34 Ω cm. Using an estimated tip diameter of 20 nm and the film thickness of 170 nm, the estimated locally measured resistance values were calculated to be 900 MΩ, which is in agreement (same order of magnitude) with local IV resistance measurements. The difference between locally measured resistance and Hall effect-based estimations of resistance may stem from contact differences between the NiO surface and the electrodes, as well as from the difference between the estimated and real contact area, suggesting that the local resistance measurements are comparable to the expected resistance, validating the spatially resolved resistance values measured by the SPM.

4 Conclusions

Spatially resolved heterogeneous surface potential and resistive response of polycrystalline NiO_x nanoparticle assembly was measured with sub-25-nm resolution under increasing level of RH in argon atmosphere. Surface potential values of NiO_x changed from 280 mV to 100 mV at 0% and 80% RH, respectively, while the average resistance decreased from 5 GΩ to 3 GΩ at 0% and 80% RH, respectively. The heterogeneity of surface potential and the spatially resolved resistance become less evident, suggesting that under high levels of RH the surface potential and resistance properties at the nanoscale and mesoscale converge. This is due to screening effect of water adsorbed on the polar NiO_x surface in the form of hydroxyl groups and molecular water.

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